

Problem set-up and explanation

Multilayer Perceptron (MLP) was used for this project to address a non-linear regression task. The project focuses on predicting the electrical energy output (PE) based on four features (exhaust vacuum (V), atmospheric pressure (AP), humidity (RH), and ambient temperature (AT)) from a Combined Cycle Power Plant (CCPP) dataset obtained from the UCI Machine Learning Repository (1). The dataset contains 9,568 instances collected from a power plant over six years (2006-2011). The energy output is influenced by the thermodynamics of turbine efficiency.

Feature	Description	Range	Unit
AT	Ambient Temperature	1.81 – 37.11	°C
V	Exhaust Vacuum	25.36 – 81.56	cm Hg
AP	Atmospheric Pressure	992.89 – 1033.30	mbar
RH	Relative Humidity	25.56 – 100.16	%
PE	Energy Output (Target)	420.26 – 495.76	MW

# AT	# V	# AP	# RH
25.24	63.47	1011.3	66.21
26.09	70.4	1007.41	85.37
26.63	73.68	1015.15	85.13
32.06	71.85	1007.9	56.44
28.7	71.64	1007.11	69.85

Figure 1- Original input data before scaling

# AT	# V	# AP	# RH
0.7480528889250169	0.7200693115916972	-0.3266001679234362	-0.4971172163033127
0.8618194799941927	1.2651572126690895	-0.9852111276742843	0.818150100611211
0.9340947260851984	1.5231497545931654	0.32523844262689516	0.8016749358899429
1.6608624784447568	1.3792087937025976	-0.9022498499678774	-1.167793713498244
1.2111498361007205	1.362690978518435	-1.0360037466781982	-0.2472438846974221

Figure 2- Input data after scaling-all data values have comparable values

The dataset was split into a 90/10 split with a fixed seed (42) to ensure reproducibility. Furthermore, the input features and output energy values were normalised (Figure 3) because neural networks (NNs) are sensitive to features with high large values, such as AP (~1000) with high magnitudes (Figure 2) would dominate features such as AT (~26). Therefore, independent scalers were used for x and y during model training to prevent data leakage.

Network architecture

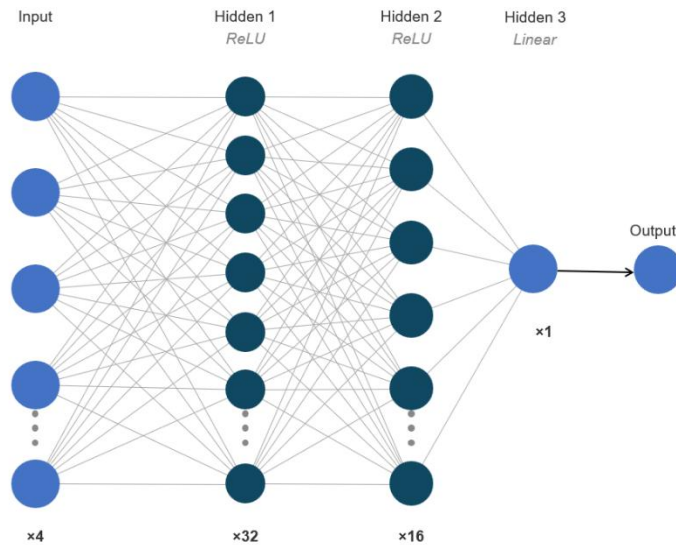


Figure 3- Architecture of the NN of 32:16:1 with ReLU activation and a linear output

MLP was chosen because it is suitable for tabular regression with continuous numerical features. Whereas, the other types of NNs such as convolutional neural networks (CNNs) are used for data that contains images and recurrent neural networks (RNNs) are used for sequential data. Therefore, MLP is the most suitable choice since the features in the CCPP dataset are independent measurements with no temporal structure or images included. The architecture was a bottleneck design ($32 \rightarrow 16 \rightarrow 1$) with 705 trainable parameters. The important components that influenced the output were: 1) ReLU activation, which avoids the gradient from vanishing; 2) linear output, which allows for the predictions to be unbounded; 3) L2 regularisation, which is used to penalise large weights; 4) Dropout, which was applied after the first hidden layer to provide stochastic regularisation and prevent overfitting; 5) Huber loss, which provides strength to the outliers in the dataset; and 6) AdamW optimiser, which dissociates weight decay from adaptive learning.

Training and training performance

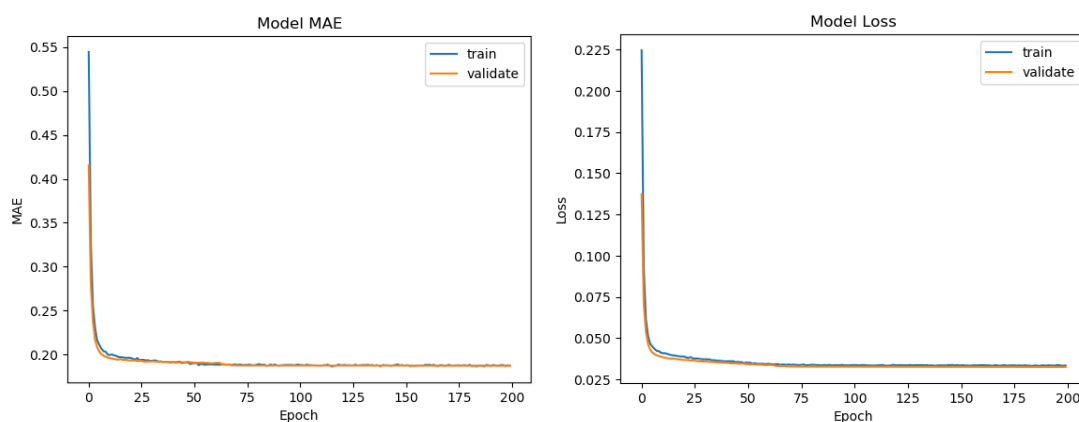


Figure 4- (A) Validation and training MAE plot for 200 epochs (B) The model Huber loss of training and validation for 200 epochs

The batch size for this model was 128, running on 200 epochs and a 10% validation split. The random seeds were set at 42 for reproducibility. One of the key observations from both plots in Figure 4 is that both plots converge rapidly from ~0.55 to 0.2 (training) and ~0.42 to 0.2 (validation) for MAE, and for the model loss, it drops from ~0.225 to 0.047 (training) and ~0.13 to 0.045 (validation). Moreover, both plots in Figure 4 have a smooth plateau and stabilise at approximately 40 epochs. Both the plots in Figure 4 show no overfitting, where both the training and validation curves converge throughout, and the validation curve is slightly lower than the training curve which is a good sign because it shows the model is able to generalise well to unseen data without memorising the training data

Evaluation, final network performance and conclusions

The trained model was evaluated using a test set of 957 data points. Three metrics were calculated: coefficient of determination (R^2), mean absolute error (MAE), and root mean square error (RMSE). The R^2 was 0.95, which implies that the model explains 95% of the variance in the energy output. The MAE was 3.10 which implies a 0.67% error compared to the 460 MW average energy output and RMSE of 3.85 MW, with a small difference between MAE and RMSE. This small difference between the RMSE and MAE indicates that many errors are similar in magnitude, except for a few outliers.

Evaluation of True vs Predicted values

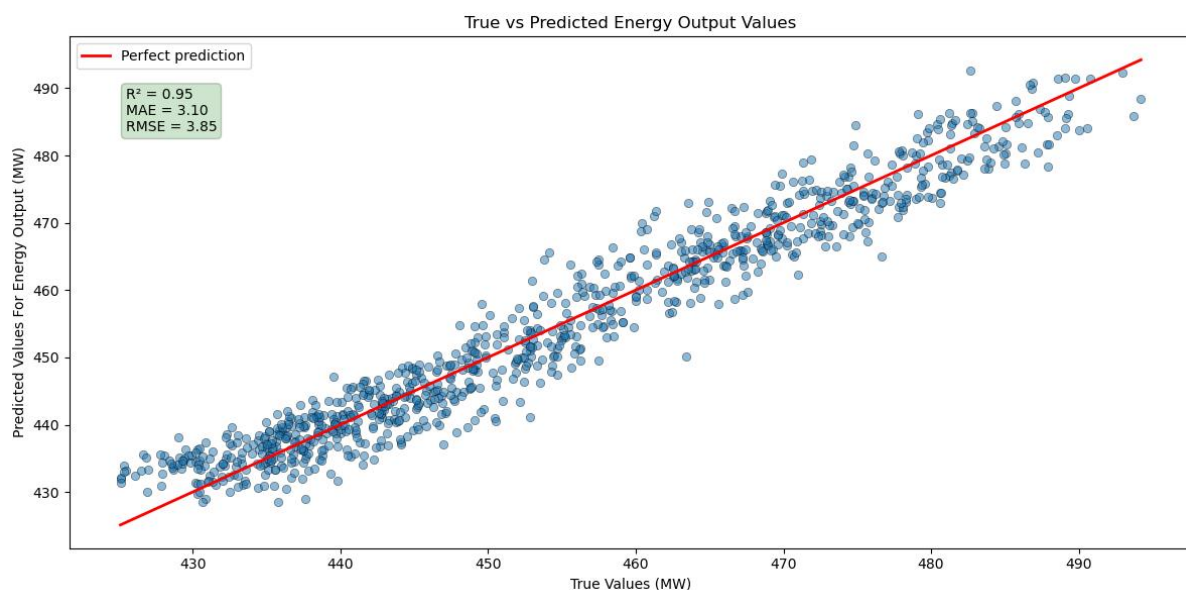


Figure 5- True vs predicted values plot with close clustering around the perfect prediction line.

The scatter plot (Figure 5) shows that most predicted values are closely aligned with the true values along the perfect prediction line with no systematic bias, which indicates that the model does not consistently over- or under-predict the energy output level.

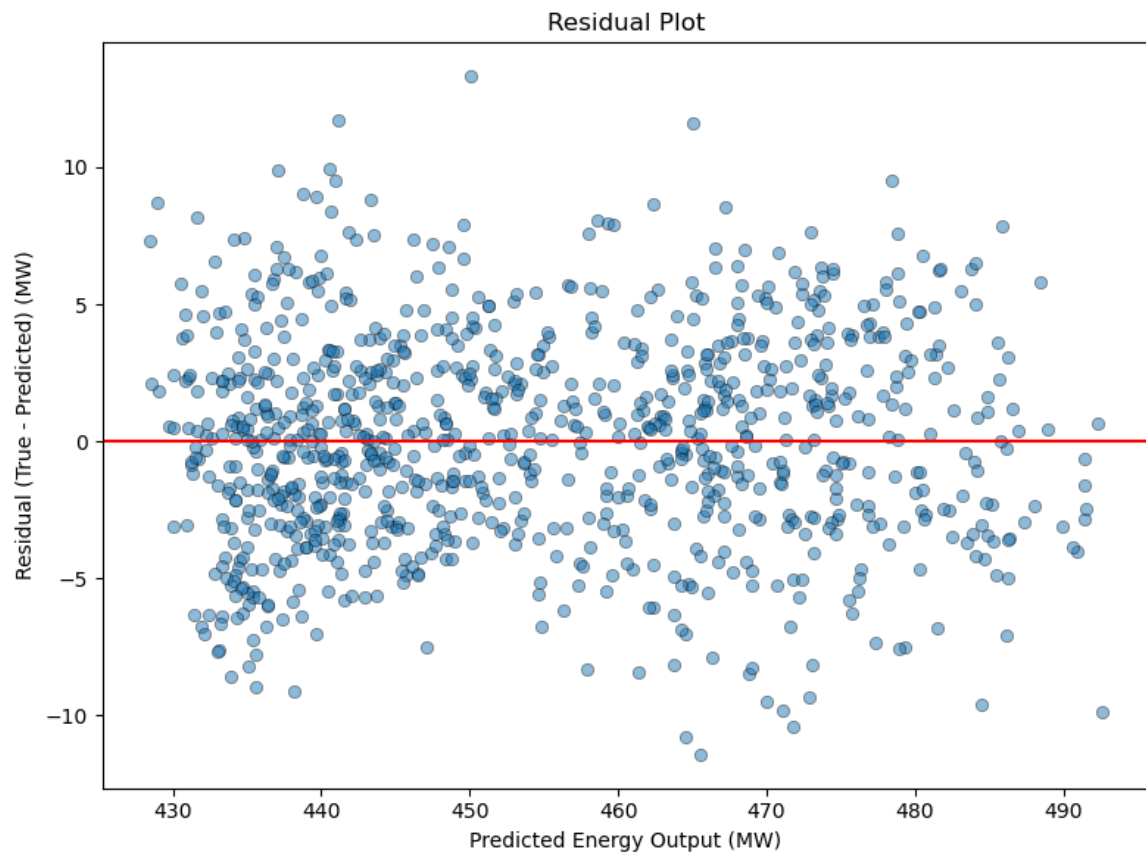


Figure 6- Residual plot of the predicted energy output

The residual plot (Figure 6) demonstrates random scatter around zero with a nearly constant variance throughout the prediction range.

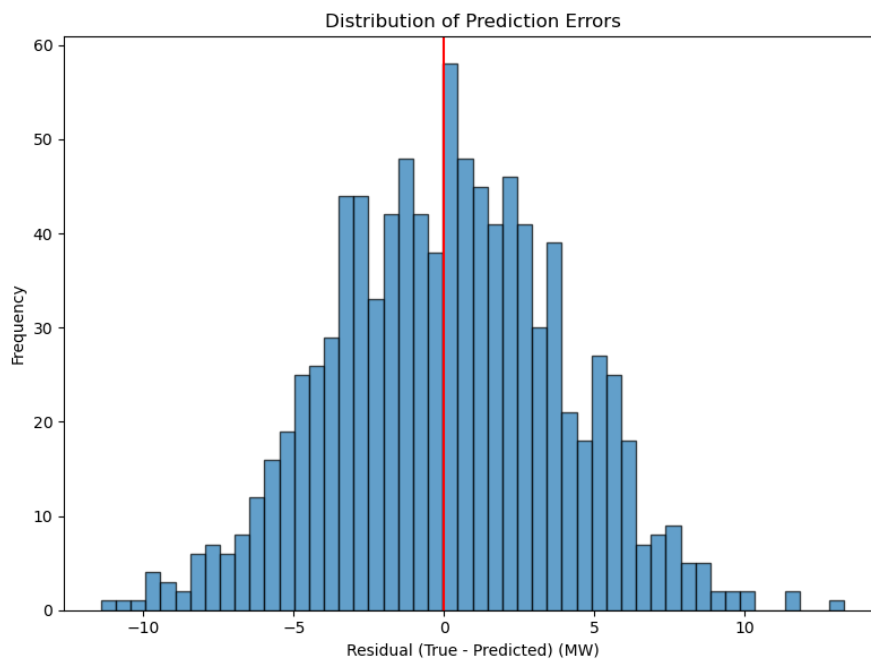


Figure 7- Distribution of prediction errors from the residual values

The histogram showing the distribution of prediction errors (Figure 7) is nearly symmetrically centred at approximately zero, which confirms the model's unbiased predictions. Most of the prediction errors lie in the ± 5 MW range, except for a few outliers that extend to ± 10 -13 MW.

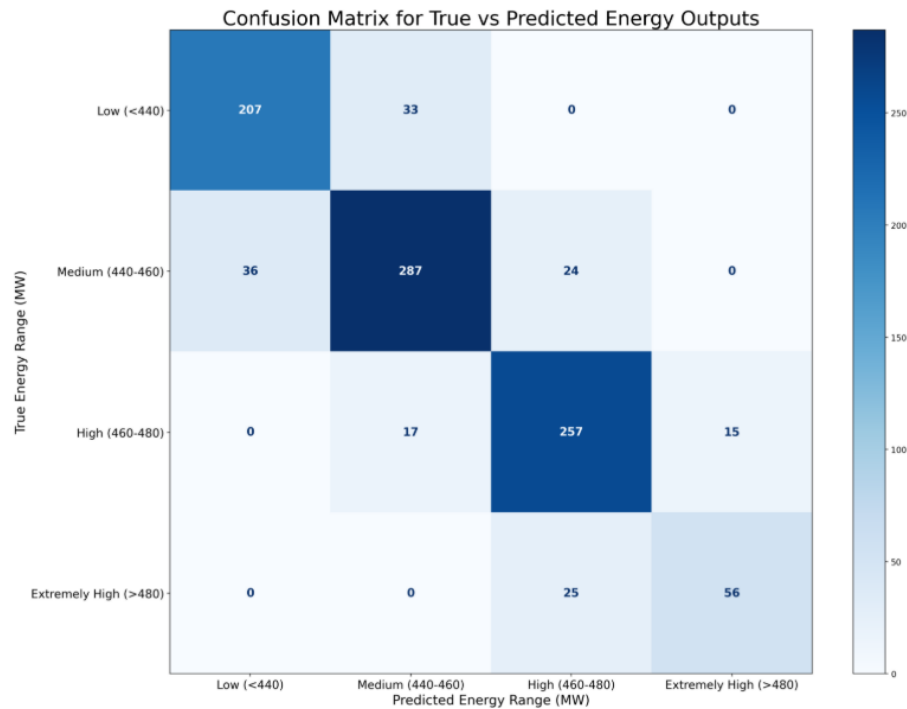


Figure 8- Confusion matrix of the true vs predicted energy outputs demonstrating a robust dominance diagonally with misclassifications only present in the nearby bins.

The confusion matrix (Figure 8) shows the performance of the number of true vs. predicted values of the energy outputs in each region. The accuracy for predicting the true energy output values for each region was as follows: low (<440) was 86.2% (207/240), Medium (440-460) was 82.7% (287/347), High (460-480) was 88.9% (257/289), and Extremely High (>480) was 69.1% (56/81). This confirms that all mispredictions occur between nearby bins, where the model does not confuse the low and high energy outputs. However, the 69.1% accuracy in the extremely high region shows that the model struggles to predict the true values with a small sample of 81 compared to the medium region with 347 samples and a much higher prediction accuracy.

Conclusion

In conclusion, this project was successful in developing an MLP model for the CCPD dataset to predict the energy output and achieve 0.95 R^2 , 3.10 MW MAE, and 3.85 MW RMSE. The use of dropout and L2 regularisation effectively prevented the model from overfitting by the convergence of the training/validation curves. The analysis of the residual plot showed that the model produced unbiased and homoscedastic predictions. However, a limitation of the model was that it could not predict values >480 MW (69.1%) owing to the lack of samples in that range, indicating an imbalance of classes. Finally, future improvements to the model could incorporate oversampling extremes, tuning hyperparameters, and comparison with gradient boosting.

References:

UCI ML Repository: <https://archive.ics.uci.edu/dataset/294/combined+cycle+power+plant>